

AIM : Naive Bayes' Classifier

- Implement the Naive Bayes' algorithm for classification.
- Train a Naive Bayes' model using a given dataset and calculate class probabilities.
- Evaluate the accuracy of the model on test data and analyze the results.

CODE :

```
# Import libraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import (
    accuracy_score,
    confusion_matrix,
    classification_report,
    roc_curve,
    auc
)

# Load dataset
print("Libraries imported")

# Read dataset
pima_df = pd.read_csv("diabetes.csv")

print("Dataset Loaded")

# Show first 5 rows
print("\nFirst 5 rows:")
print(pima_df.head())

# Select input and output
X = pima_df.drop("Outcome", axis=1)
y = pima_df["Outcome"]
```

Name : Shubham Sanjay Karape

Roll No : T085

SHETH L.U.J. AND SIR M.V. COLLEGE OF ARTS, SCIENCE AND COMMERCE

SUBJECT : ARTIFICIAL INTELLIGENCE

```
# Split data into training and testing
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.30,
    random_state=7,
    stratify=y
)
```

```
# Scale the data
scaler = StandardScaler()
```

```
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

```
print("\nData Transformed")
```

```
# Create Naive Bayes model
model = GaussianNB()
```

```
# Train the model
model.fit(X_train, y_train)
```

```
print("\nModel:")
print(model)
```

```
# Predict test data
predicted = model.predict(X_test)
```

```
print("\nPredicted Values:")
print(predicted)
```

```
# Check model accuracy
accuracy = accuracy_score(y_test, predicted)
```

```
print("\nAccuracy:")
print(accuracy)
```

Name : Shubham Sanjay Karape
Roll No : T085

SHETH L.U.J. AND SIR M.V. COLLEGE OF ARTS, SCIENCE AND COMMERCE

SUBJECT : ARTIFICIAL INTELLIGENCE

```
print("\nAccuracy Percentage:")
print(f"{accuracy * 100:.2f}%")

# Create confusion matrix
cm = confusion_matrix(y_test, predicted)

print("\nConfusion Matrix:")
print(cm)

# Show classification report
print("\nClassification Report:")
print(classification_report(y_test, predicted))

# Get prediction probability
y_predict_prob = model.predict_proba(X_test)[: , 1]

# Calculate ROC curve
fpr, tpr, thresholds = roc_curve(
    y_test,
    y_predict_prob
)

# Calculate AUC score
roc_auc = auc(fpr, tpr)

print("\nROC AUC Score:")
print(roc_auc)

# Plot ROC curve
plt.figure(figsize=(8, 6))

plt.plot(
    fpr,
    tpr,
    color='darkorange',
    label='ROC curve (AUC = %0.2f)' % roc_auc
```

Name : Shubham Sanjay Karape

Roll No : T085

SHETH L.U.J. AND SIR M.V. COLLEGE OF ARTS, SCIENCE AND COMMERCE

SUBJECT : ARTIFICIAL INTELLIGENCE

)

```
# Draw reference line
```

```
plt.plot(  
    [0, 1],  
    [0, 1],  
    color='navy',  
    linestyle='--'  
)
```

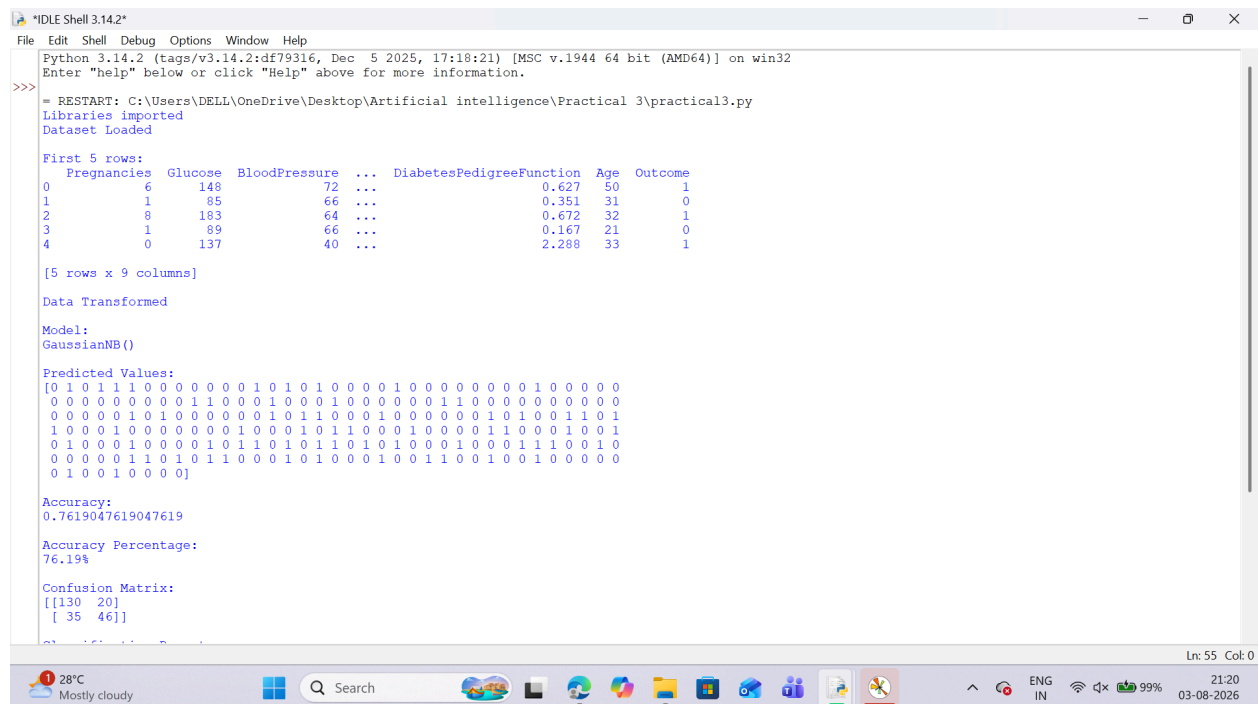
```
# Add graph details
```

```
plt.xlabel('False Positive Rate')  
plt.ylabel('True Positive Rate')  
plt.title('ROC Curve - Gaussian Naive Bayes')  
plt.legend(loc="lower right")  
plt.grid()
```

```
# Show graph
```

```
plt.show()
```

OUTPUT :



```
*IDLE Shell 3.14.2*  
File Edit Shell Debug Options Window Help  
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)] on win32  
Enter "help" below or click "Help" above for more information.  
>>>  
= RESTART: C:\Users\DELL\OneDrive\Desktop\Artificial intelligence\Practical 3\practical3.py  
Libraries imported  
Dataset Loaded  
  
First 5 rows:  
Pregnancies Glucose BloodPressure ... DiabetesPedigreeFunction Age Outcome  
0 6 148 72 ... 0.627 50 1  
1 1 85 66 ... 0.351 31 0  
2 8 183 64 ... 0.672 32 1  
3 1 89 66 ... 0.167 21 0  
4 0 137 40 ... 2.288 33 1  
  
[5 rows x 9 columns]  
  
Data Transformed  
  
Model:  
GaussianNB()  
  
Predicted Values:  
[0 1 0 1 1 1 0 0 0 0 0 0 0 1 0 1 0 1 0 0 0 0 0 1 0 0 0 0 0 0 0 1 0 0 0 0 0 0  
0 0 0 0 0 0 0 0 0 0 1 1 0 0 0 1 0 0 0 1 0 0 0 0 0 0 0 1 1 0 0 0 0 0 0 0 0 0 0  
0 0 0 0 0 1 0 1 0 0 0 0 0 0 0 1 0 1 1 0 0 0 1 0 0 0 0 0 0 0 1 0 1 0 0 1 1 0 1  
1 0 0 0 1 0 0 0 0 0 0 0 1 0 0 0 1 0 1 1 0 0 0 1 0 0 0 0 1 1 0 0 0 0 1 0 0 1  
0 1 0 0 0 1 0 0 0 0 1 0 1 1 0 1 0 1 1 0 1 0 1 0 0 0 1 0 0 0 1 1 1 0 0 1 0  
0 0 0 0 0 1 1 0 1 0 1 1 0 0 0 1 0 1 0 0 0 1 0 0 0 1 0 0 0 1 1 0 0 0 0 0  
0 1 0 0 1 0 0 0 0]  
  
Accuracy:  
0.7619047619047619  
  
Accuracy Percentage:  
76.19%  
  
Confusion Matrix:  
[[130 20]  
 [ 35 46]]  
  
Ln: 55 Col: 0
```

Name : Shubham Sanjay Karape

Roll No : T085

SHETH L.U.J. AND SIR M.V. COLLEGE OF ARTS, SCIENCE AND COMMERCE
SUBJECT : ARTIFICIAL INTELLIGENCE

```
*IDLE Shell 3.14.2*
File Edit Shell Debug Options Window Help

[5 rows x 9 columns]

Data Transformed

Model:
GaussianNB()

Predicted Values:
[0 1 0 1 1 1 0 0 0 0 0 0 0 1 0 1 0 1 0 0 0 0 0 1 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0
0 0 0 0 0 0 0 0 0 0 1 1 0 0 0 0 1 0 0 0 0 1 0 0 0 0 0 0 0 1 1 0 0 0 0 0 0 0 0 0 0 0
0 0 0 0 0 1 0 1 0 0 0 0 0 0 0 1 0 1 1 0 0 0 0 1 0 0 0 0 0 0 0 1 0 1 0 0 0 1 1 0 1
1 0 0 0 1 0 0 0 0 0 0 0 1 0 0 0 1 0 1 1 0 0 0 1 0 0 0 0 0 1 1 0 0 0 0 1 0 0 1
0 1 0 0 0 1 0 0 0 0 1 0 1 1 0 1 0 1 1 0 1 0 1 0 0 0 1 0 0 0 0 1 1 1 0 0 0 1 0
0 0 0 0 0 1 1 0 1 0 1 1 0 0 0 1 0 1 0 0 0 1 0 0 0 1 1 0 0 0 1 1 1 0 0 0 1 0
0 1 0 0 1 0 0 0 0]

Accuracy:
0.7619047619047619

Accuracy Percentage:
76.19%

Confusion Matrix:
[[130  20]
 [ 35  46]]

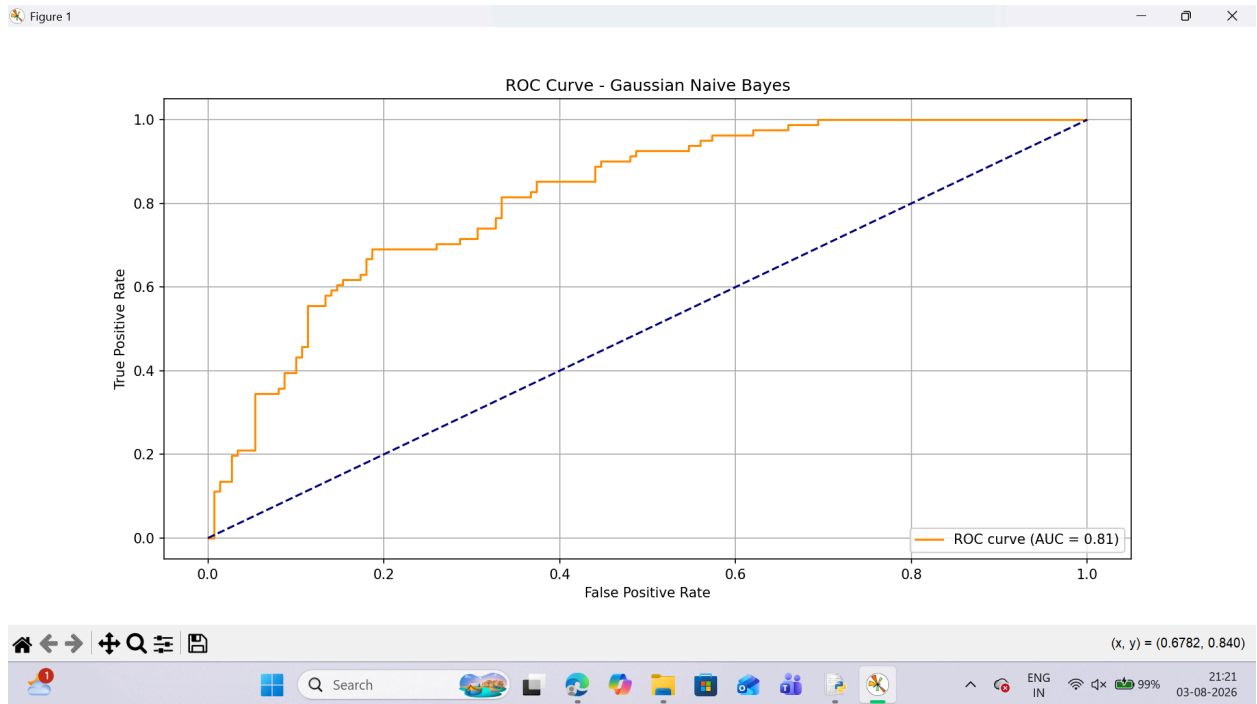
Classification Report:
      precision    recall  f1-score   support

     0       0.79       0.87       0.83       150
     1       0.70       0.57       0.63        81

   accuracy       0.74       0.72       0.76       231
  macro avg       0.74       0.72       0.73       231
 weighted avg       0.76       0.76       0.76       231

ROC AUC Score:
0.8143209876543209

Ln: 55 Col: 0
```



Name : Shubham Sanjay Karape
Roll No : T085